

Data Exploration

Team1

Index

Test data

- Dataset
- Point out Data issues
- Interesting insight
- Summarize
- What we learning

Dataset

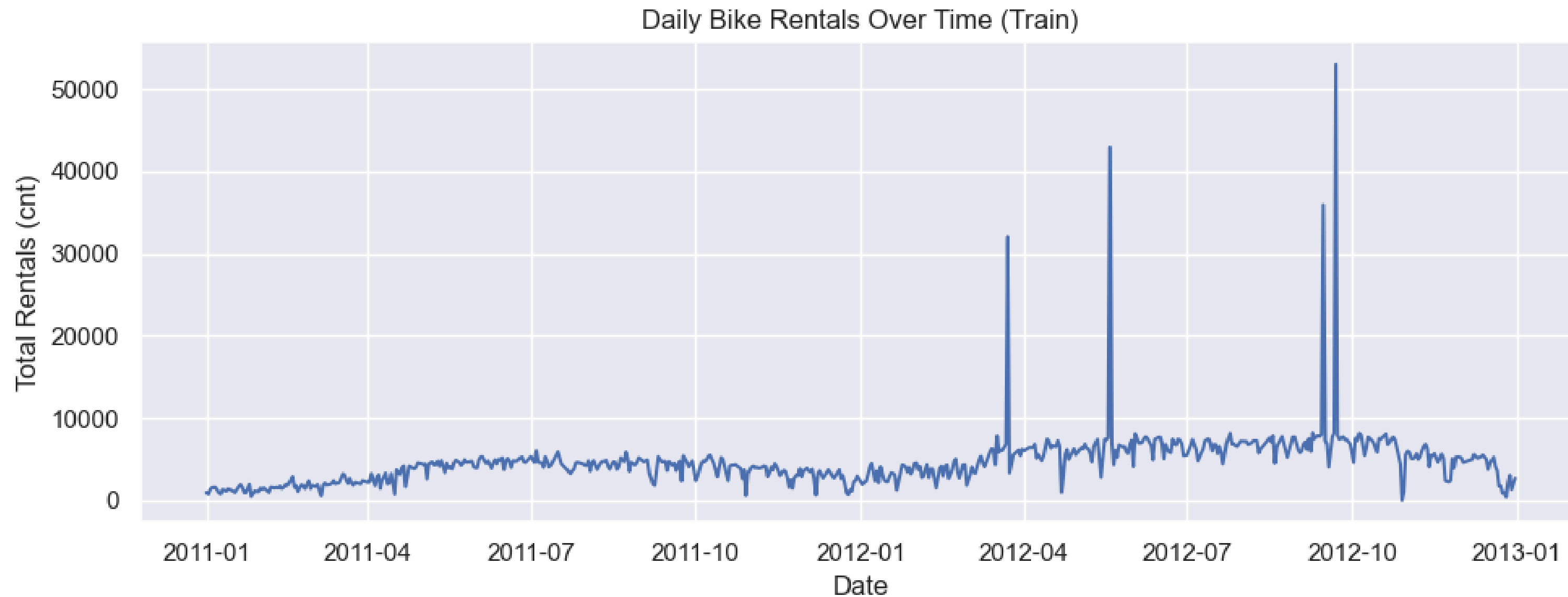
Descriptive Statistics

- Format: 731 rows × 18 columns
- 15 features
- 3 labels (casual, registered, cnt)

	instant	dteday	season	yr	mnth	holiday	weekday	workingday	weathersit	temp	atemp	hum	windspeed	leaflets	price reduction	casual	registered	cnt
0	299	26.10.2011	4.0	0	10	0	3	1	2	19.3667	0.472846	108.06255	0.148642	605	0	404	3490	3894
1	458	02.04.2012	2.0	1	4	0	1	1	1	17.3565	0.427513	75.65220	0.312139	518	0	1208	4728	5936
2	687	17.11.2012	4.0	1	11	0	6	0	1	13.0000	0.326383	81.81255	0.179729	766	0	1313	4316	5629
3	346	12.12.2011	4.0	0	12	0	-1	1	1	9.5333	0.270196	NaN	0.063450	739	0	143	3167	3310
4	291	18.10.2011	4.0	0	10	0	2	1	2	21.3000	0.522721	105.25005	0.110087	463	0	637	4111	4748
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
727	78	19.03.2011	1.0	0	3	0	6	0	1	18.9000	0.466525	56.87505	0.368167	1022	1	1424	1693	3117
728	81	22.03.2011	NaN	0	3	0	2	1	1	17.6667	0.440642	93.68745	0.225750	551	0	460	2243	2703
729	377	12.01.2012	1.0	1	1	0	4	1	2	15.3000	0.381938	120.43755	0.180967	520	0	269	3828	4097
730	299	26.10.2011	4.0	0	10	0	3	1	2	19.3667	0.472846	108.06255	0.148642	605	0	404	3490	3894
731	497	11.05.2012	2.0	1	5	0	5	1	1	21.3333	0.520833	54.06255	0.236937	570	1	1319	5711	7030

Dataset

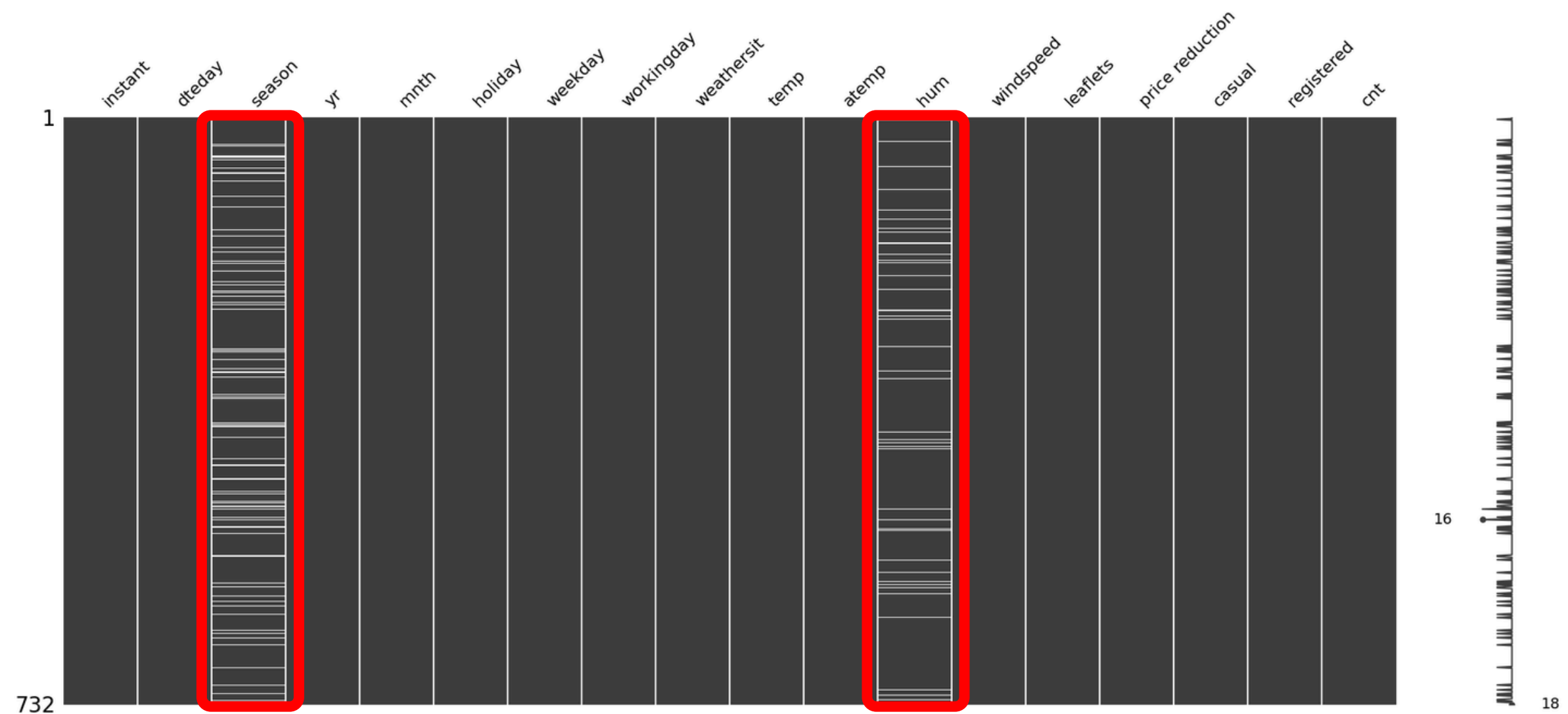
Descriptive Statistics



Point out Data issues

Missing Values

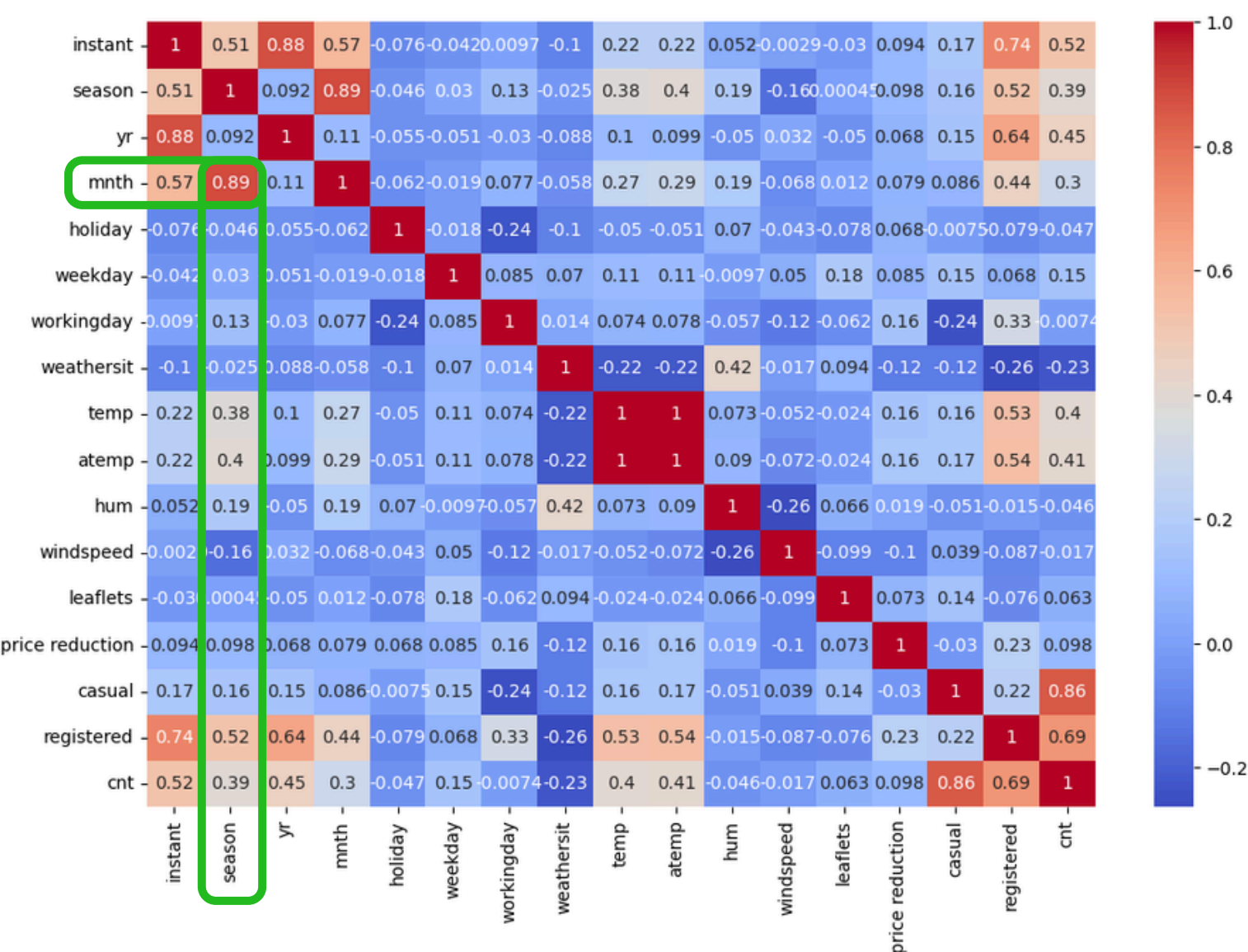
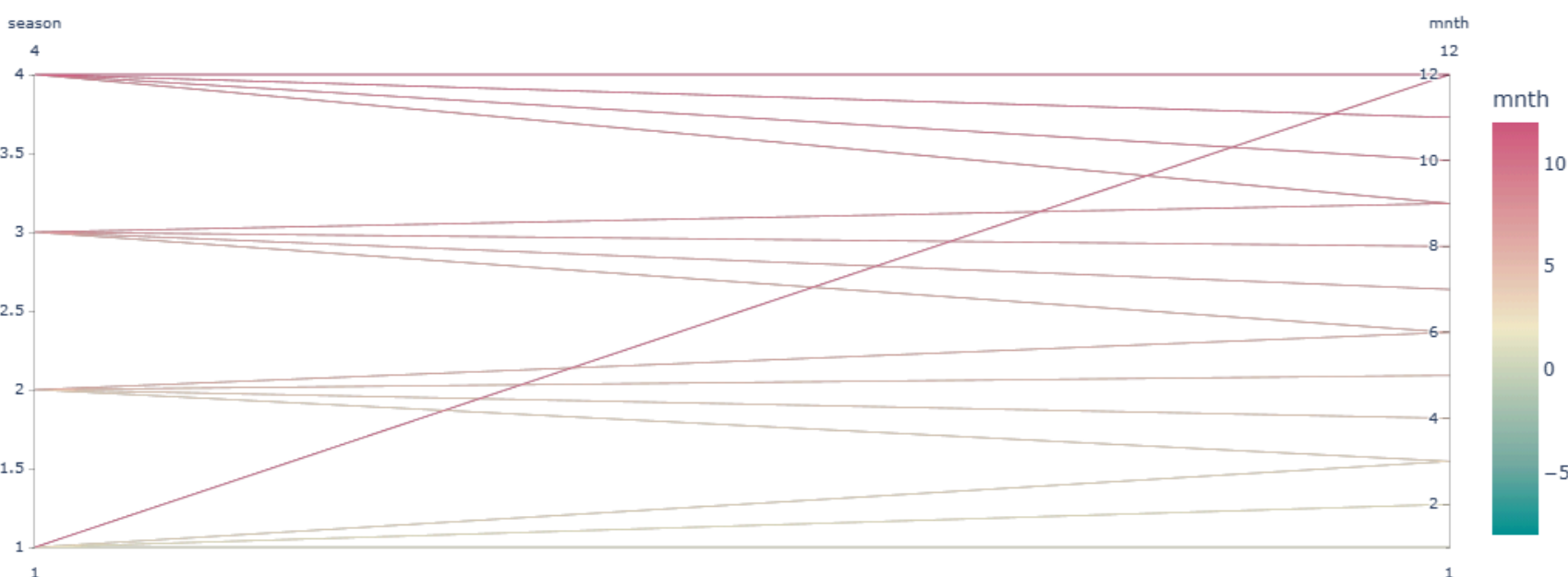
- season and humidity(hum) have missing value



Point out Data issues

Missing Values

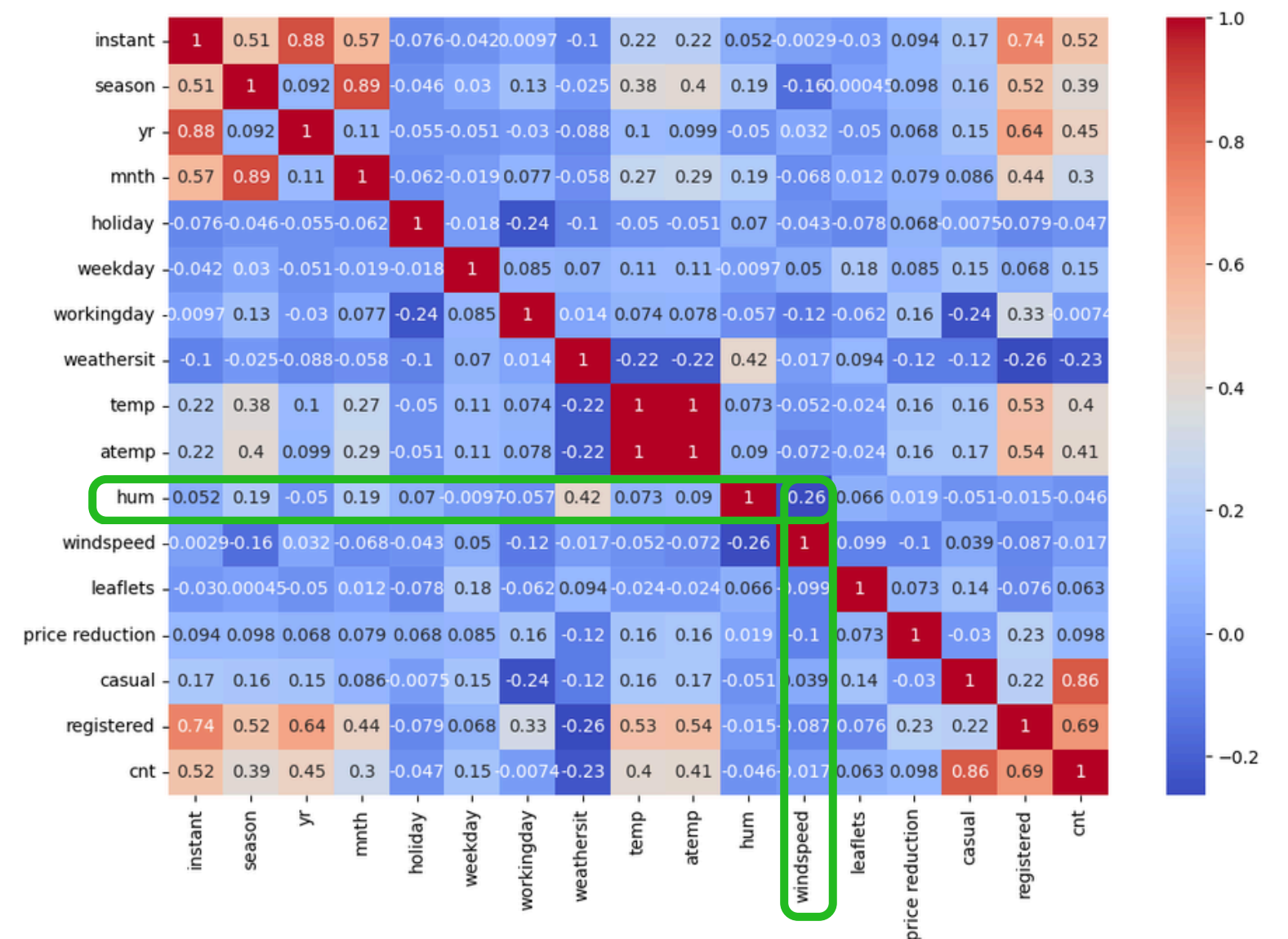
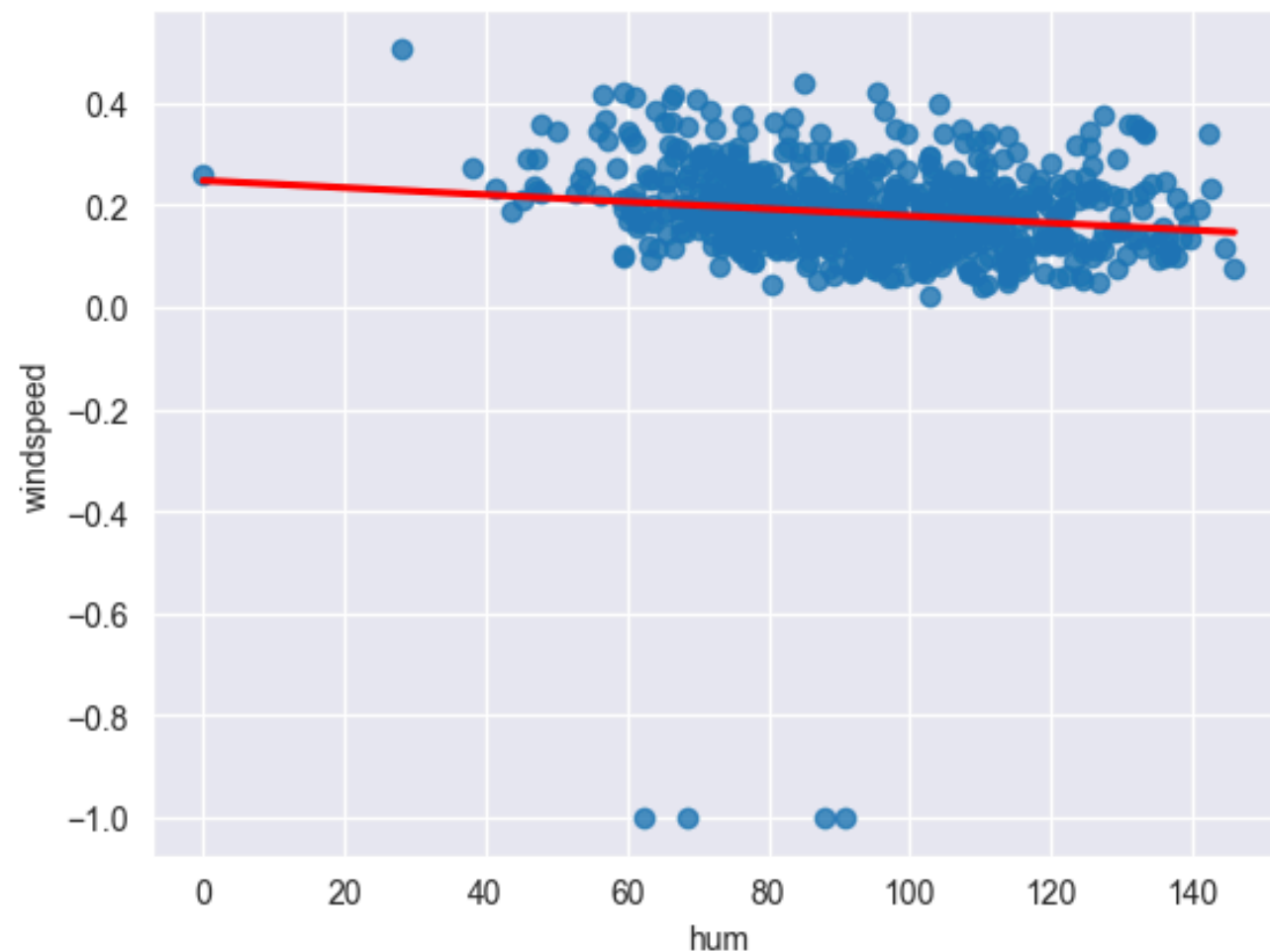
- Season and month correlate to each other → could be used for rule based imputation



Point out Data issues

Missing Values

- hum and windspeed are correlated → could be used for regression imputation; otherwise: average imputation



Point out Data issues

False Values

- weekday: false value (-1), should be 0-6 → Correct using dteday (rule-based imputation) or remove the row.
- negative windspeed (-1) is impossible → Replace with median windspeed or remove the row.
- hum max is > 100 (more than 100 % is impossible) → Find out what scale is used. If normalized or in percent: cap values at 100 (Winsorisation) or mark/remove them as outliers

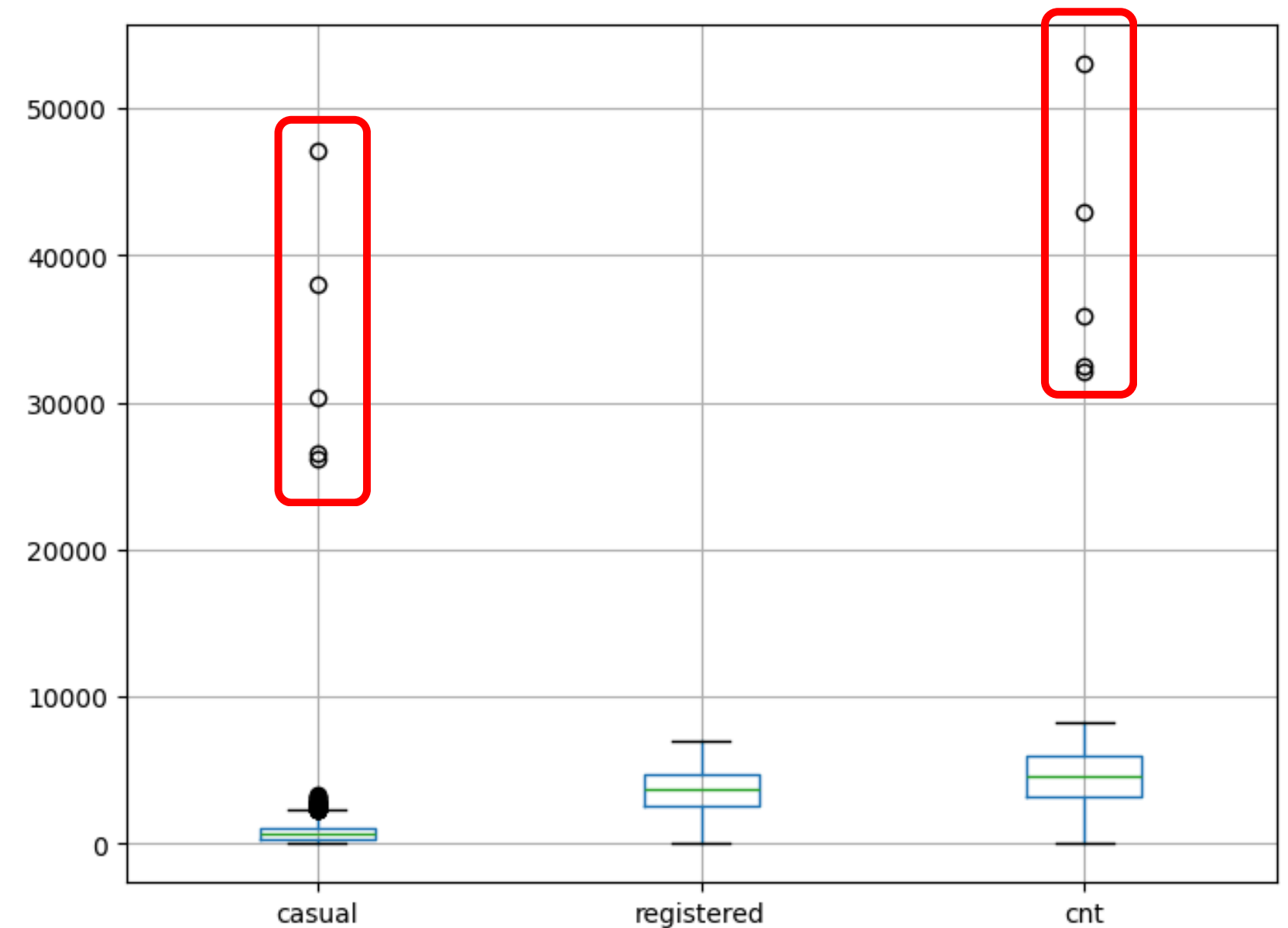
	instant	season	yr	mnth	holiday	weekday	workingday	weathersit	temp	atemp	hum	windspeed	leaflets	price reduction	casual	registered	cnt
count	732.000000	659.000000	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000	692.000000	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000
mean	365.908470	2.476480	0.500000	6.524590	0.028689	2.867486	0.684426	1.396175	19.814779	0.474352	93.808823	0.183994	741.775956	0.153005	1058.079235	3655.945355	4714.024590
std	211.035856	1.112505	0.500342	3.451948	0.167044	2.095577	0.465062	0.544980	7.317049	0.162850	21.381071	0.117027	166.967863	0.360239	2862.520234	1559.200902	3506.692931
min	1.000000	1.000000	0.000000	1.000000	0.000000	-1.000000	0.000000	1.000000	2.365200	0.079070	0.000000	-1.000000	458.000000	0.000000	2.000000	20.000000	22.000000
25%	183.750000	1.000000	0.000000	4.000000	0.000000	1.000000	0.000000	1.000000	13.491675	0.337891	77.841075	0.134341	598.000000	0.000000	315.750000	2499.000000	3157.500000
50%	365.500000	2.000000	0.500000	7.000000	0.000000	3.000000	1.000000	1.000000	19.933300	0.485158	93.350550	0.179727	739.500000	0.000000	711.500000	3660.000000	4544.500000
75%	548.250000	3.000000	1.000000	10.000000	0.000000	5.000000	1.000000	2.000000	26.208325	0.608289	109.390650	0.233205	887.000000	0.000000	1095.500000	4769.750000	5946.000000
max	731.000000	4.000000	1.000000	12.000000	1.000000	6.000000	1.000000	3.000000	34.466700	0.840896	145.875000	0.507463	1032.000000	1.000000	47138.000000	6946.000000	53021.000000

Also interesting: weatherset doesn't have values in category 4 (heavy rain)

Point out Data issues

Outliers

- casual, cnt have false data/outliers
 - ($\text{cnt} = \text{casual} + \text{registered}$, so false value makes sense)
- Use winsorisation (limit extreme values to 95%-quantile).



Point out Data issues

Features to be transformed or removed

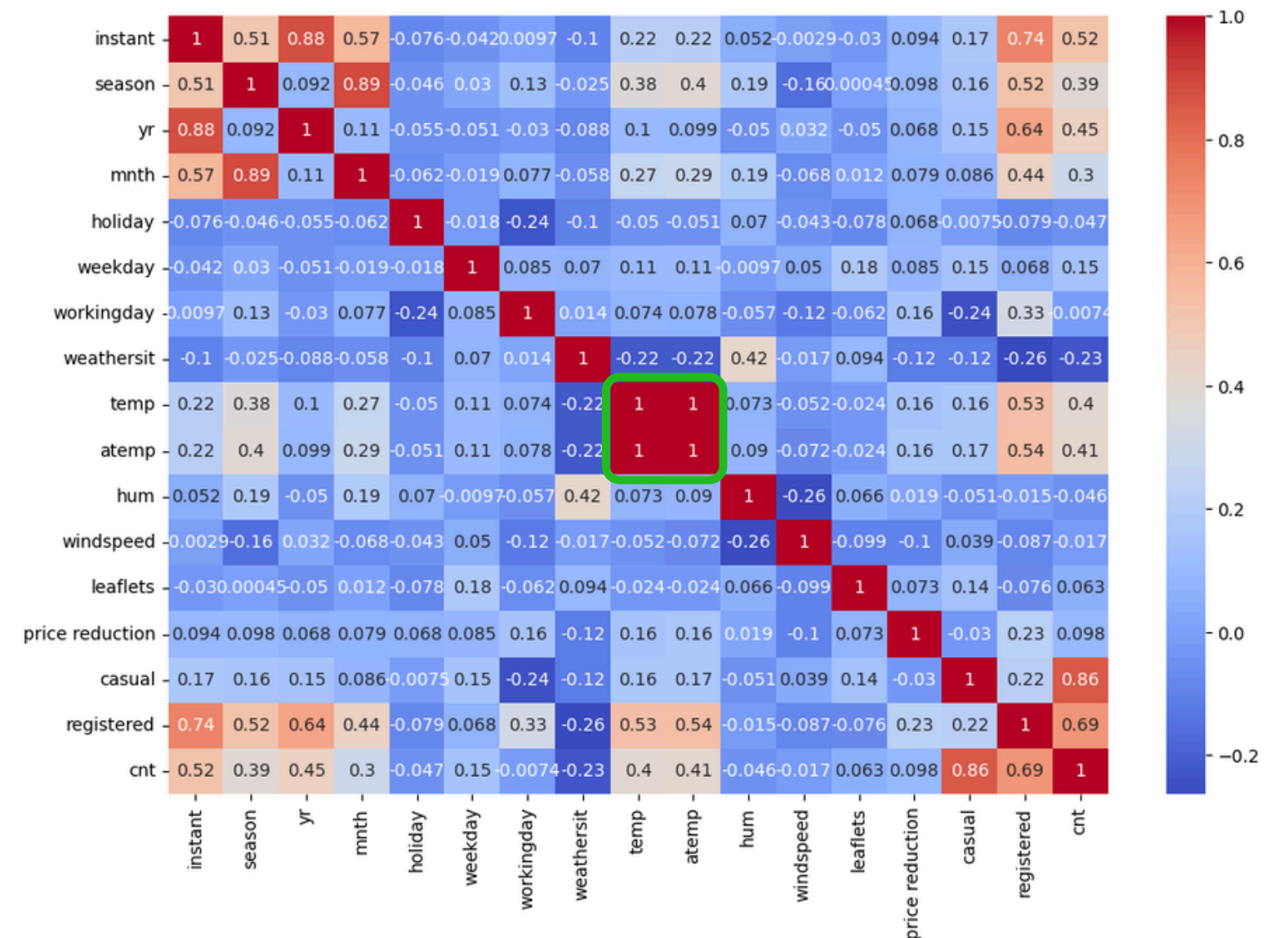
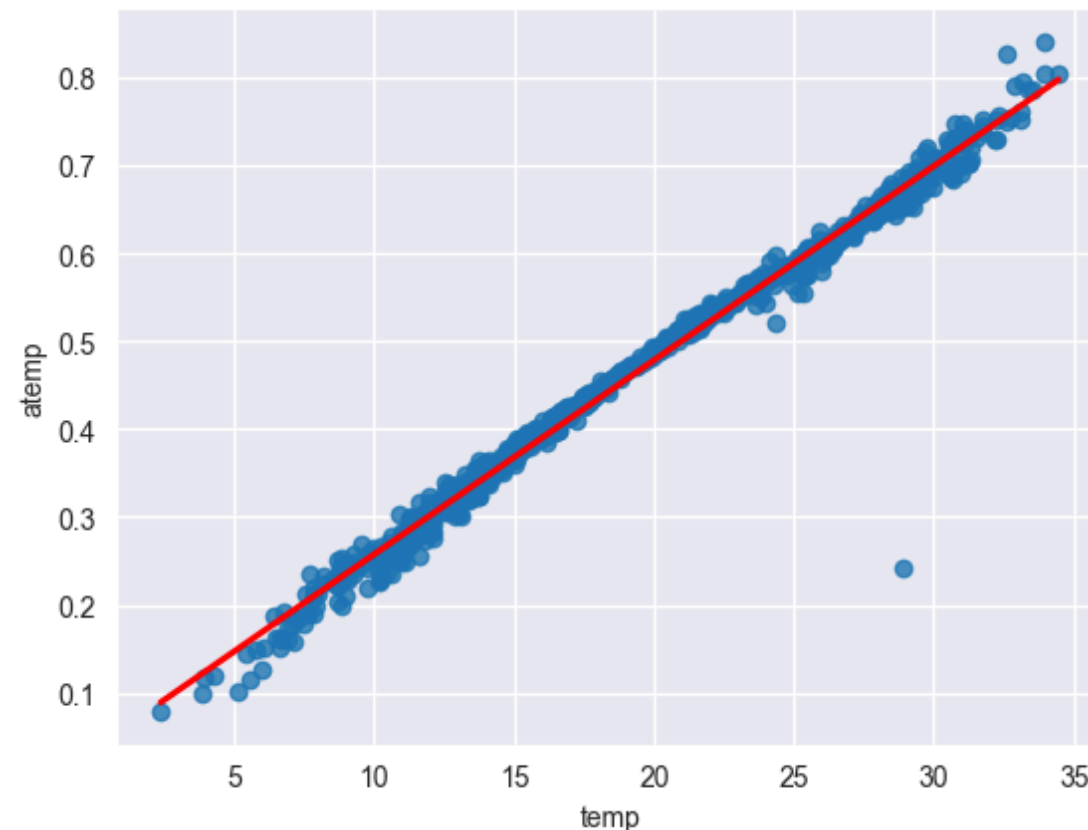
- Delete or change data format in dteday columns
- might damage machine learning
- dteday could be used to derivate missing season values and false weekday values but needs to be transformed first

	instant	dteday	season	yr	mnth	holiday	weekday	workingday	weathersit	temp	atemp	hum	windspeed	leaflets	price reduction	casual	registered	cnt
0	299	26.10.2011	4.0	0	10	0	3	1	2	19.3667	0.472846	108.06255	0.148642	605	0	404	3490	3894
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...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
727	78	19.03.2011	1.0	0	3	0	6	0	1	18.9000	0.466525	56.87505	0.368167	1022	1	1424	1693	3117
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Point out Data issues

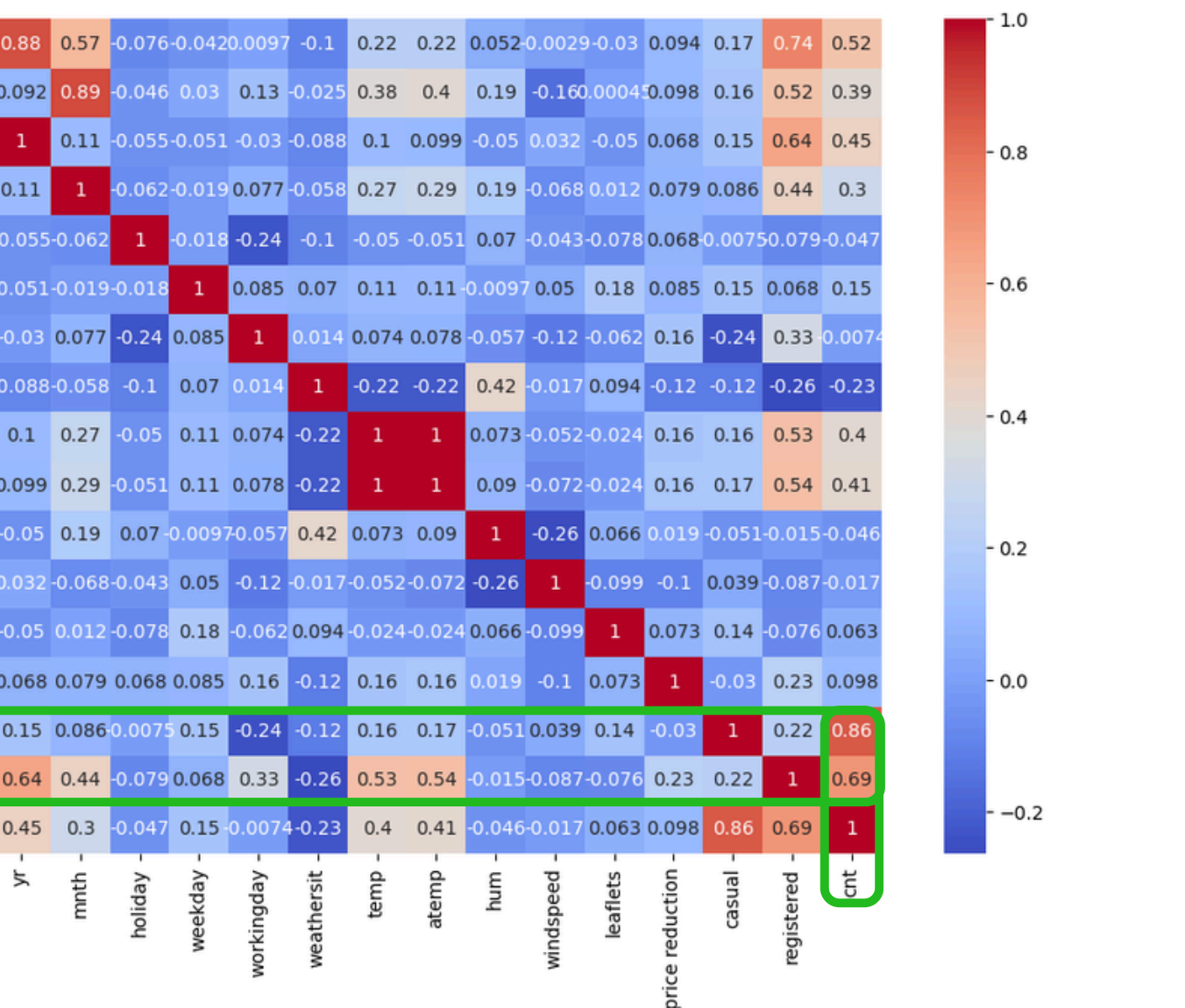
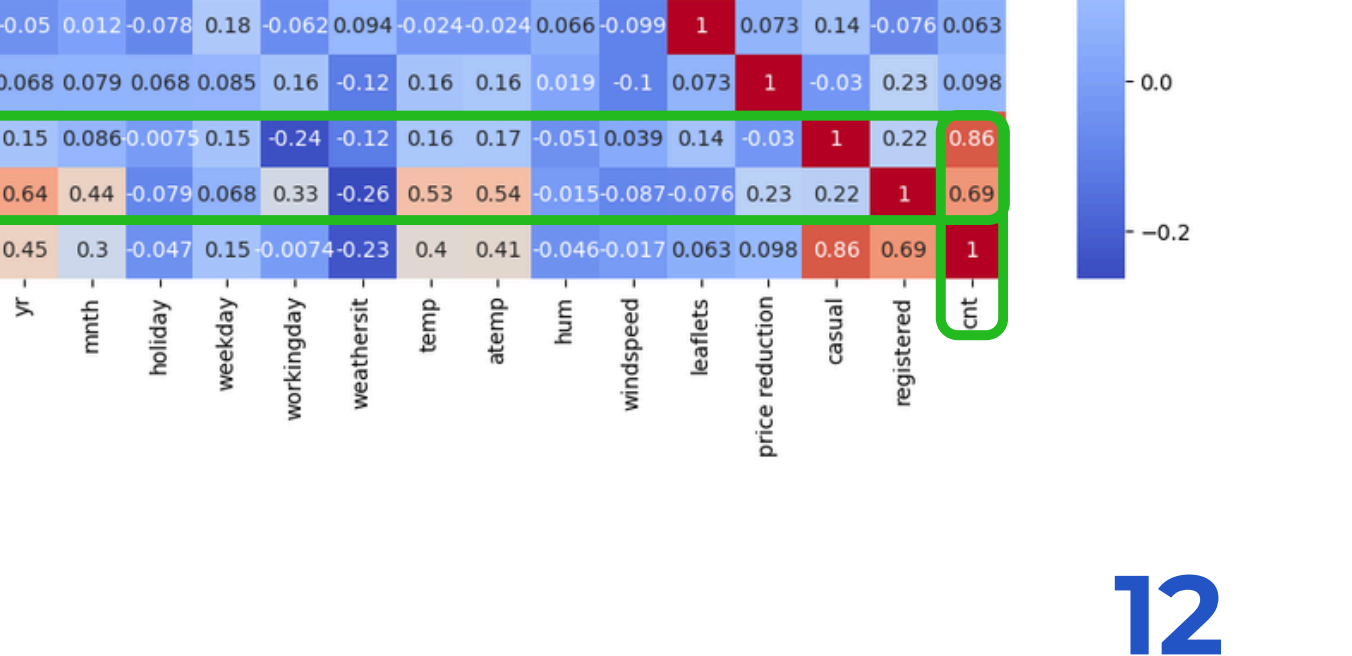
Features to be transformed or removed

- temp and atemp correlate to each other
- temp and atemp have same correlation with all other features; atemp is already normalized, so temp can be removed



- | | 0.092 | 0.89 | -0.046 | 0.03 | 0.13 | -0.025 | 0.38 | 0.4 | 0.19 | -0.16 | 0.004 | -0.098 | 0.16 | 0.52 | 0.39 |
|-------|--------|--------|--------|-------|--------|--------|--------|--------|--------|--------|-------|--------|-------|--------|------|
| 1 | 0.11 | -0.055 | -0.051 | -0.03 | -0.088 | 0.1 | 0.099 | -0.05 | 0.032 | -0.05 | 0.068 | 0.15 | 0.64 | 0.45 | |
| 0.11 | 1 | -0.062 | -0.019 | 0.077 | -0.058 | 0.27 | 0.29 | 0.19 | -0.068 | 0.012 | 0.079 | 0.086 | 0.44 | 0.3 | |
| 0.055 | -0.062 | 1 | -0.018 | -0.24 | -0.1 | -0.05 | -0.051 | 0.07 | -0.043 | -0.078 | 0.068 | 0.0075 | 0.079 | -0.047 | |
| 0.051 | 0.018 | 0.018 | 1 | 0.005 | 0.07 | 0.11 | 0.11 | 0.0037 | 0.05 | 0.10 | 0.005 | 0.15 | 0.068 | 0.15 | |

-0.03	0.077	-0.24	0.085	1	0.014	0.074	0.078	-0.057	-0.12	-0.062	0.16	-0.24	0.33	-0.0074
0.088	-0.058	-0.1	0.07	0.014	1	-0.22	-0.22	0.42	-0.017	0.094	-0.12	-0.12	-0.26	-0.23
0.1	0.27	-0.05	0.11	0.074	-0.22	1	1	0.073	-0.052	-0.024	0.16	0.16	0.53	0.4
0.099	0.29	-0.051	0.11	0.078	-0.22	1	1	0.09	-0.072	-0.024	0.16	0.17	0.54	0.41
-0.05	0.19	0.07	-0.0097	0.057	0.42	0.073	0.09	1	-0.26	0.066	0.019	-0.051	-0.015	-0.046

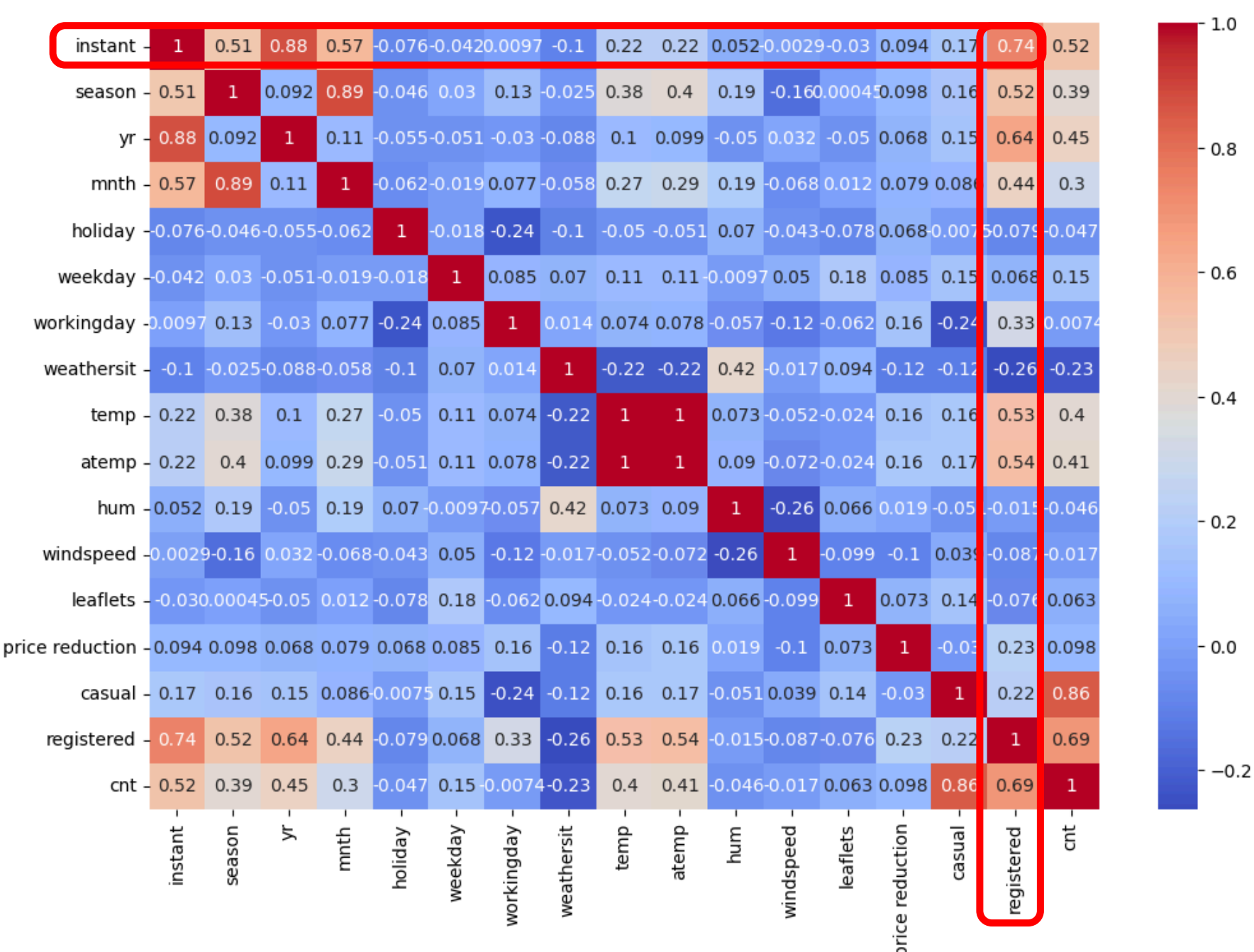
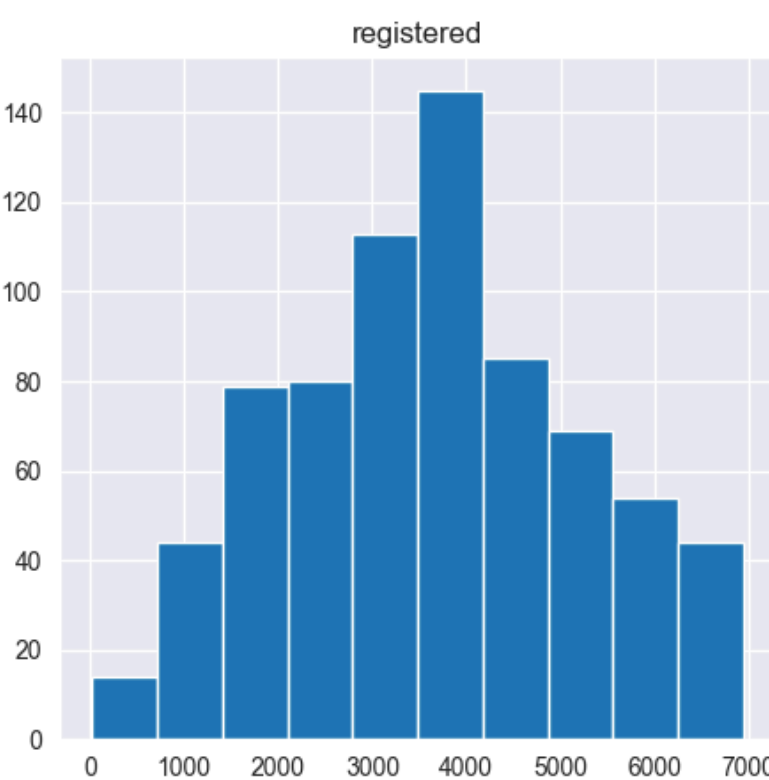
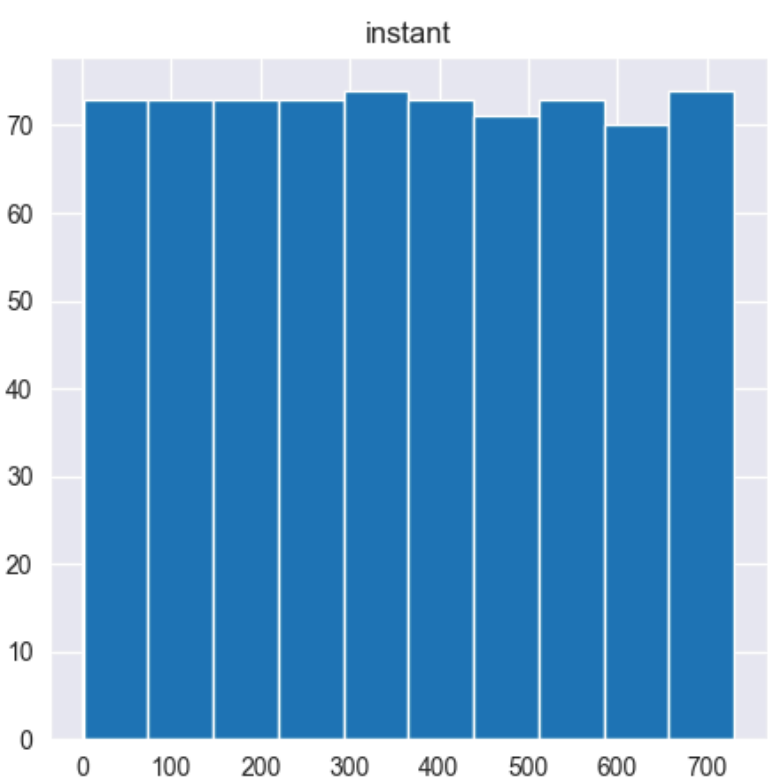


Point out Data issues

Correlation error

- instant correlate with registered
- instant shouldn't be correlate with any data

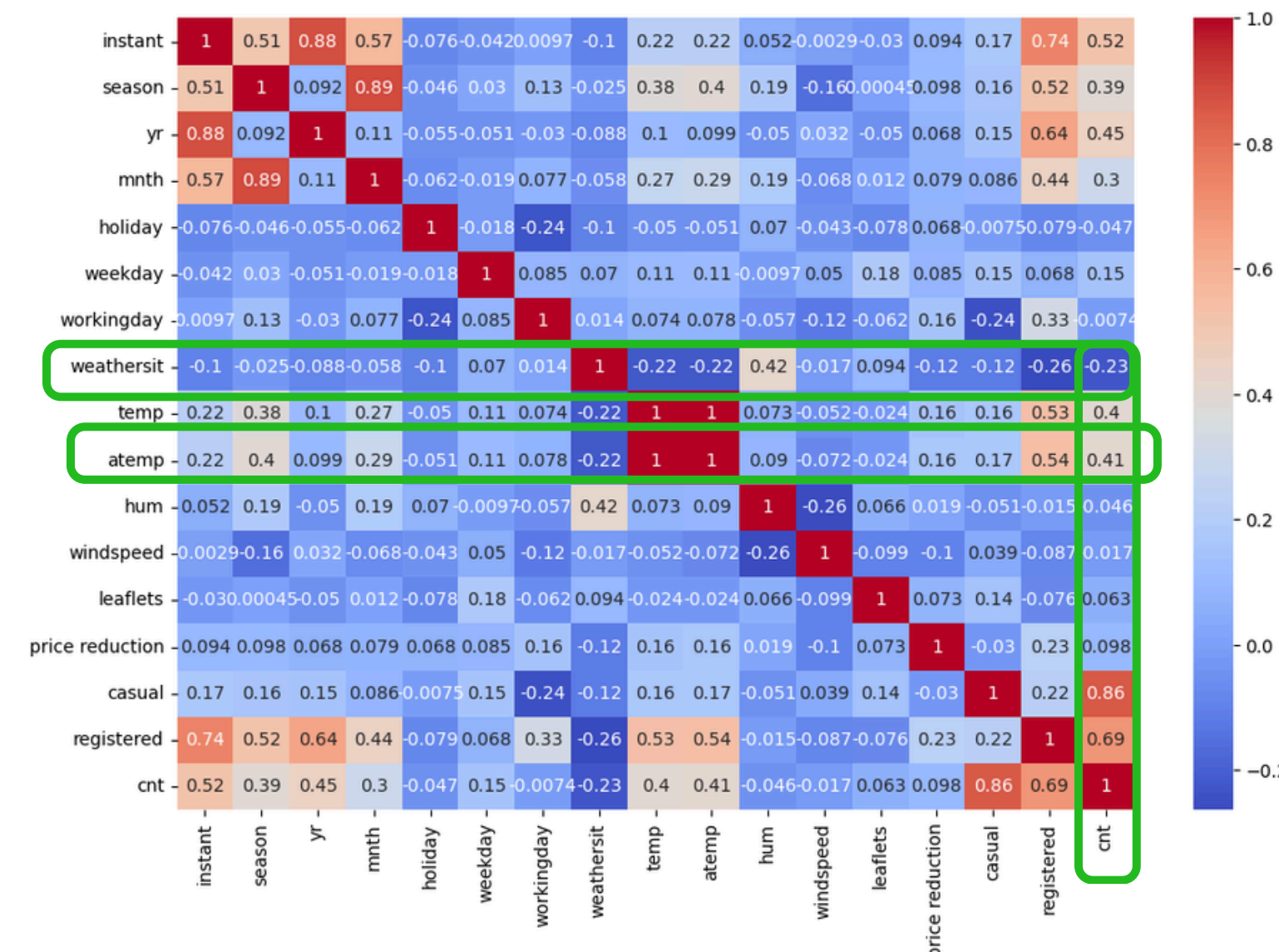
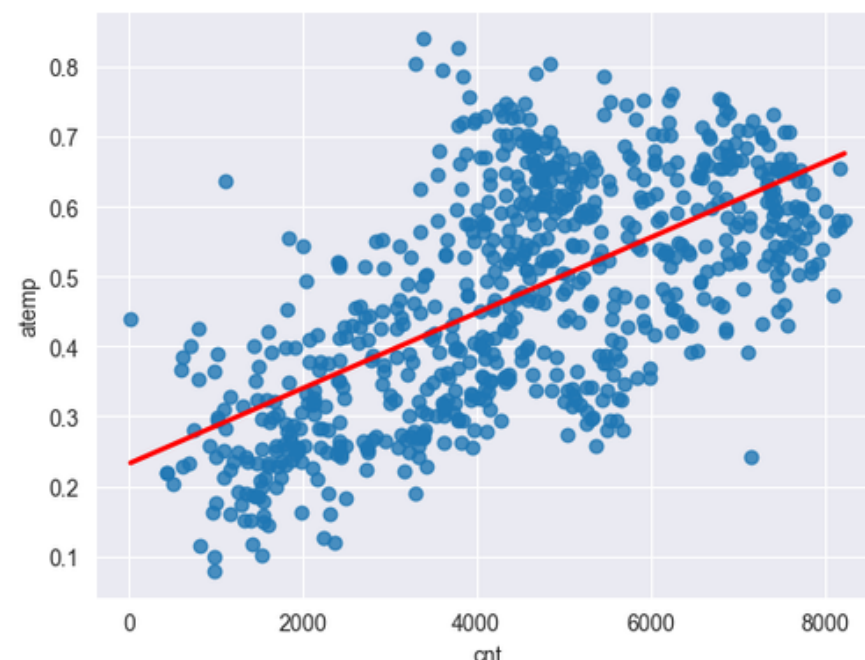
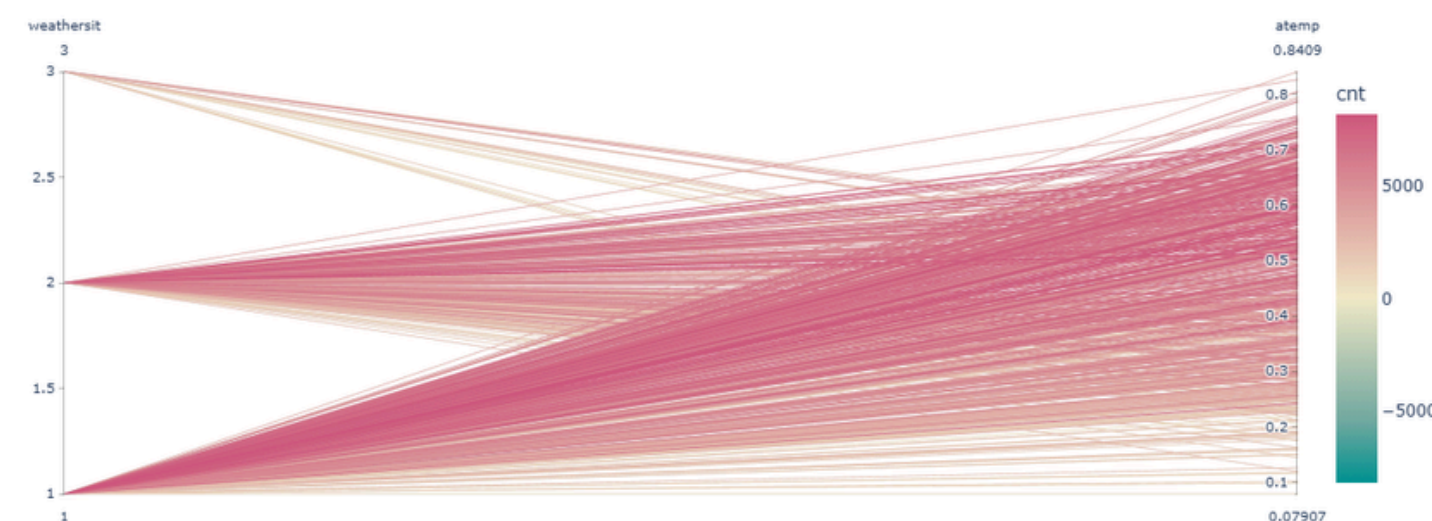
→ remove instant: only an ID, no predictive meaning



Interesting insight

Correlation

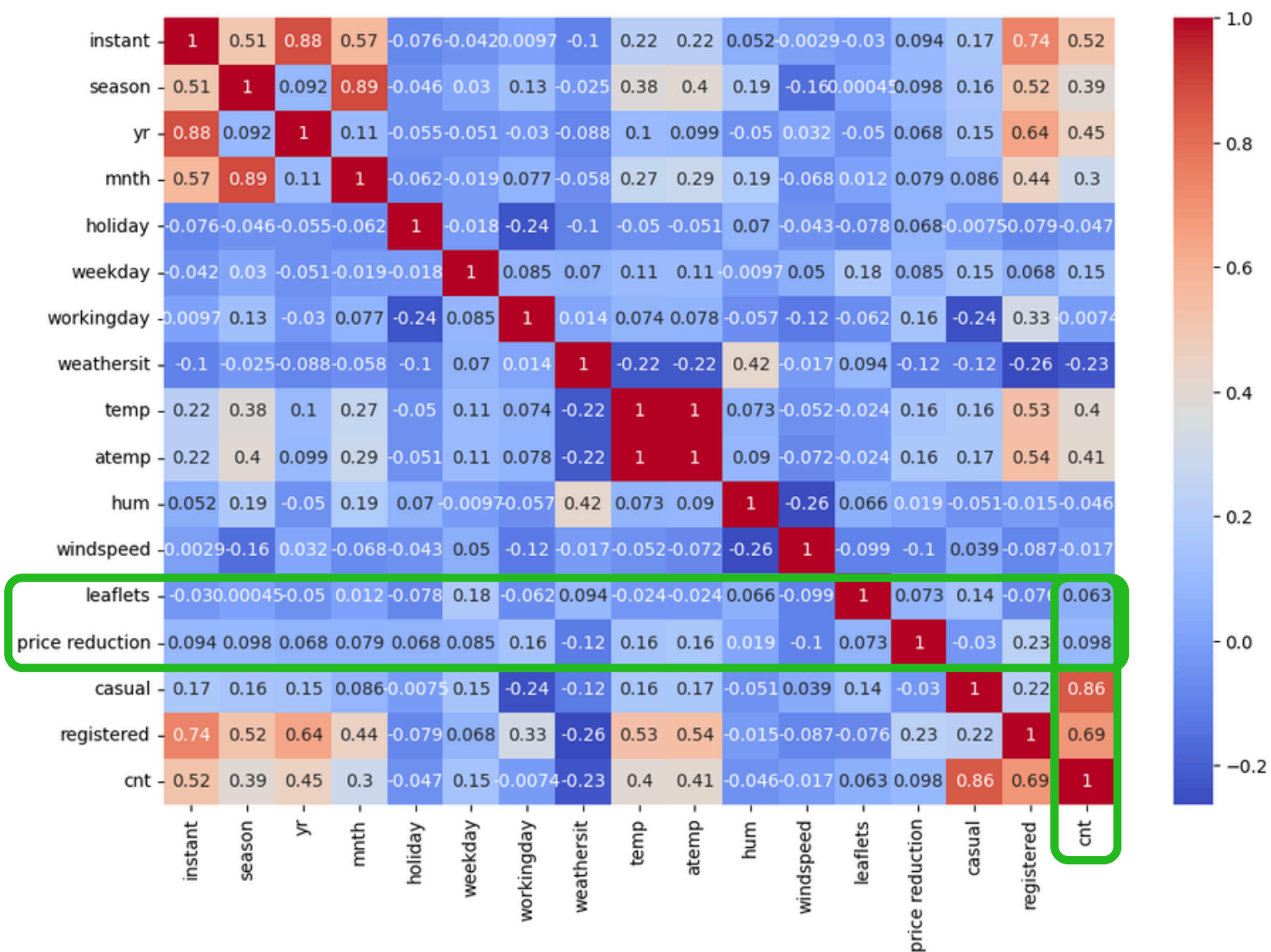
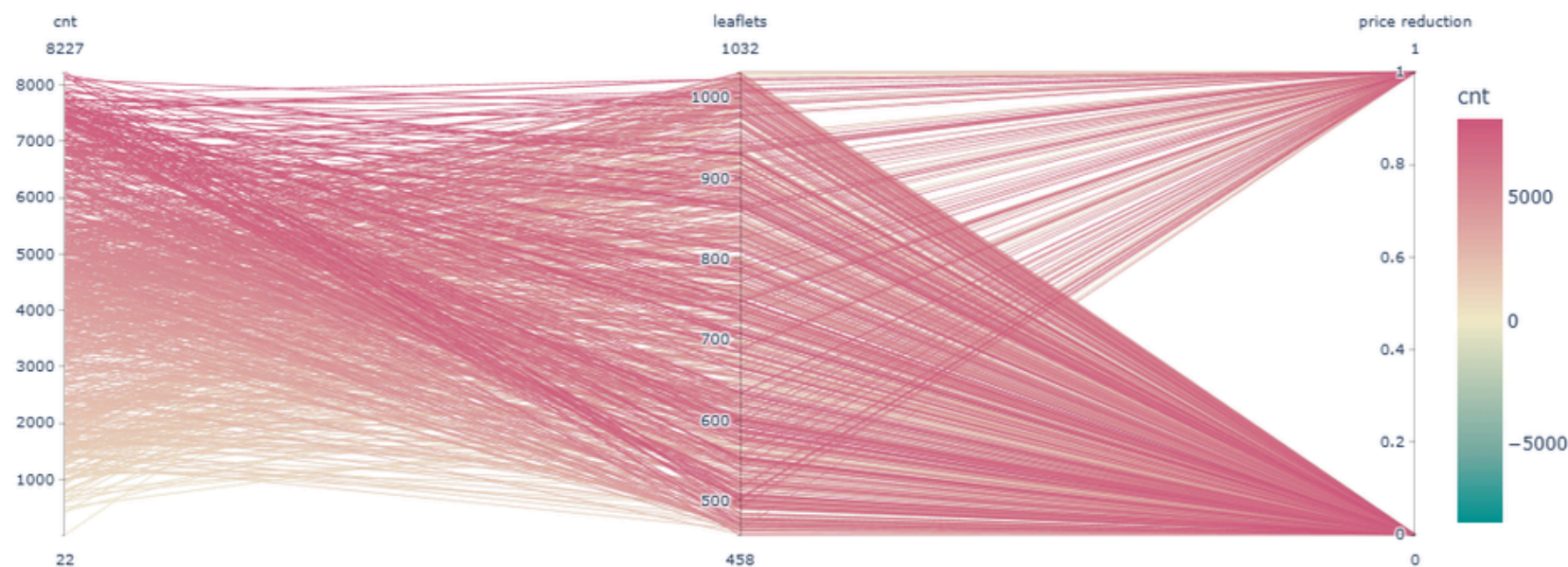
- cnt correlation with atemp and weathersit



Interesting insight

Correlation

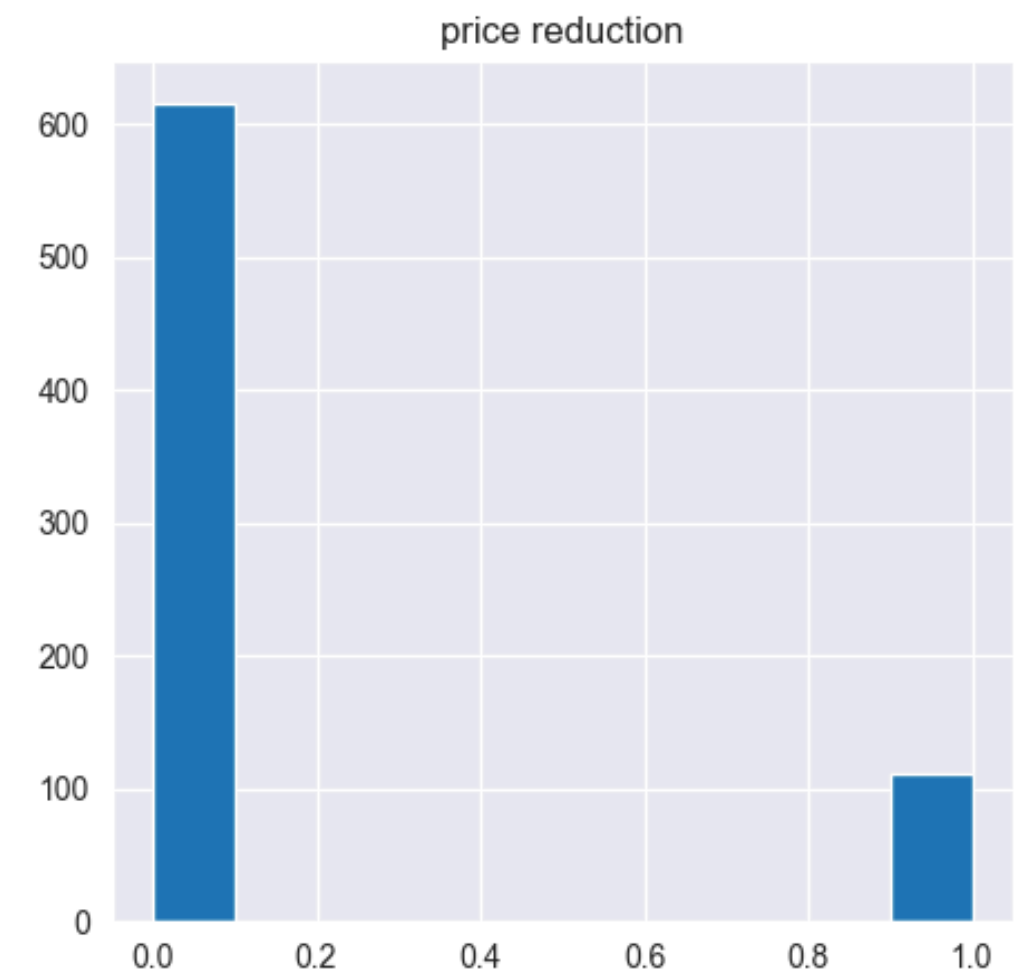
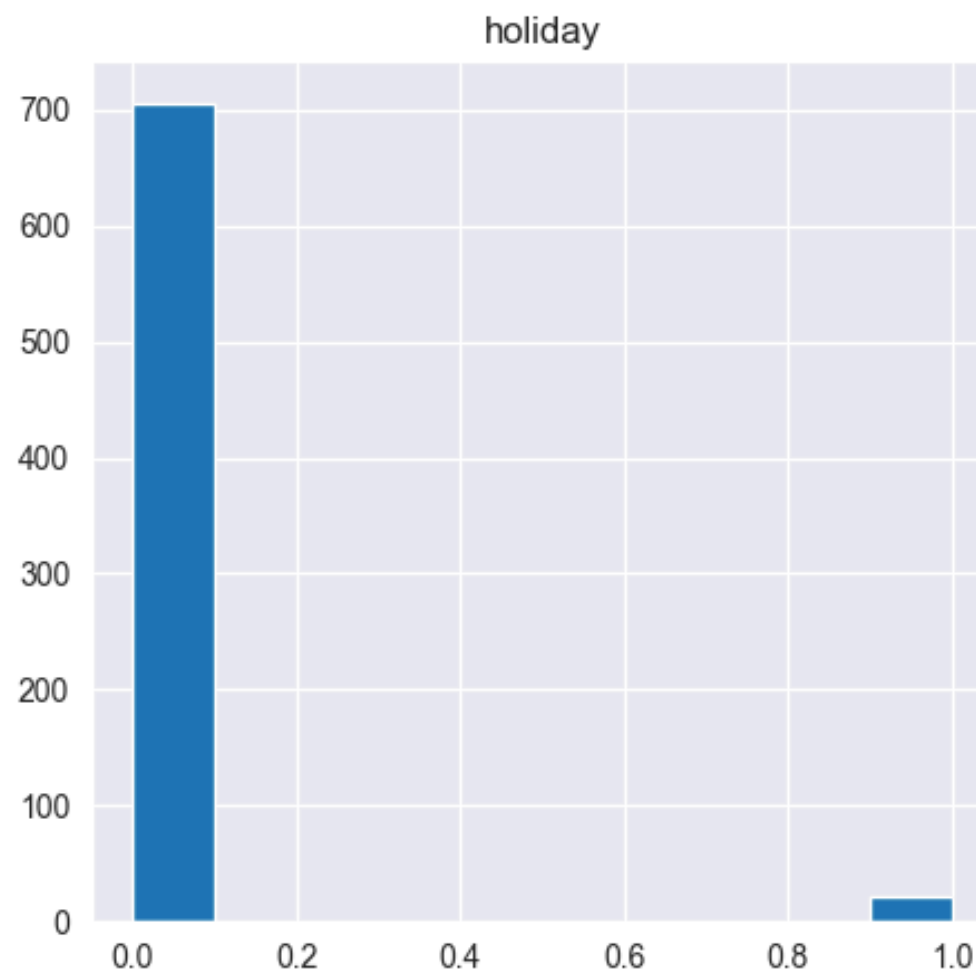
- leaflets and price reduction **didn't correlate** to cnt



Interesting insight

Histogram

- have only few user in Holiday and price reduction days



Summarize

Point out Data issues

- Missing values: season and humidity
- Outliers: weatherset, humidity (> 100), weekday (-1) casual and cnt
- Features to be transformed: Date (Change format or delete), instant (might damage machine learning)
- Features to be removed: Temp (duplicate with aTemp), Date

Interesting insight

- temp and atemp correlate to each other
- season close to month
- casual, registered correlation to cnt
- cnt correlation with atemp and weathersit
- humidity close to windspeed
- have only few user in Holiday and price reduction days
- leaflets and price reduction **didn't correlation** to cnt

What we learning

Team

- Understanding dataset structures
- Communication about problem, what can improved
- Coding in general
- Data visualization and how to present data

Myself

- Still need to communicating with team more
- Stop underestimate task, do it in last minute even I can do it on time
- reduce my perfectionist